# Advanced Statistical Analysis of Anxiety and Depression Trends During the COVID-19 Pandemic

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## Abstract

This study applies advanced statistical techniques to examine trends in self-reported anxiety and depression symptoms during the COVID-19 pandemic using the CDC Household Pulse Survey. We use generalized linear and additive models, mixed effects models, time series methods, multivariate techniques like PCA and MANOVA, and clustering to understand how mental health patterns evolved over time and across demographic subgroups.


## Introduction

The COVID-19 pandemic has profoundly impacted public mental health, leading to increased levels of anxiety and depression. To assess these trends, the U.S. Census Bureau and National Center for Health Statistics launched the Household Pulse Survey. This analysis uses advanced statistical modeling to explore temporal dynamics and subgroup differences in mental health outcomes based on this large-scale repeated cross-sectional dataset.

## Data Description

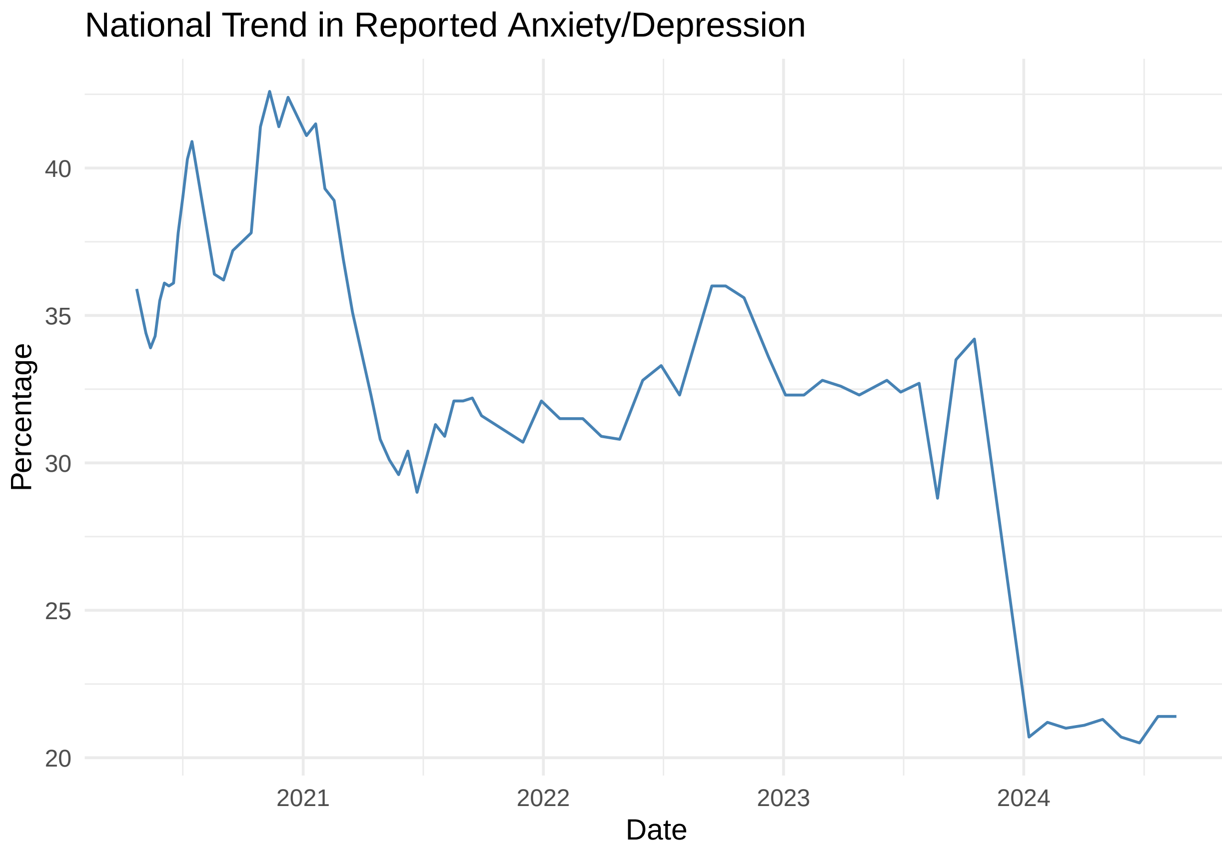
```r
setwd("C:/Users/O&1/OneDrive/Documents/R-projects/Covid")
df_raw <- read.csv("Indicators_of_Anxiety_or_Depression.csv")
```

```r
df <- df_raw %>%
  rename_with(~ gsub(" ", "_", .x)) %>%
  rename(
    indicator = Indicator,
    group = Group,
    subgroup = Subgroup,
    value = Value,
    low_ci = LowCI,
    high_ci = HighCI,
    week = Phase,
    start_date = TimePeriodStartDate,
    end_date = TimePeriodEndDate
  ) %>%
  mutate(
    start_date = mdy(start_date),  # fix here
    end_date = mdy(end_date),      # and here
    time_numeric = as.numeric(start_date)
  ) %>%
```

```r
  filter(indicator == "Symptoms of Anxiety Disorder or Depressive Disorder") %>%
  drop_na(value)
```

## Exploratory Analysis

```r
ggplot(df %>% filter(group == "National Estimate"), aes(x = start_date, y = value)) +
  geom_line(color = "steelblue") +
  labs(title = "National Trend in Reported Anxiety/Depression", x = "Date", y = "Percentage") +
  theme_minimal()
```



# Generalized Linear Model (GLM)

```
glm_model <- glm(value ~ time_numeric, data = df %>% filter(group == "National Estimate"), family = gau
summary(glm_model)
```

```
##
## Call:
## glm(formula = value ~ time_numeric, family = gaussian(), data = df %>%
##     filter(group == "National Estimate"))
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.010e+02  1.717e+01   11.70  < 2e-16 ***
## time_numeric -8.847e-03  9.037e-04   -9.79  9.6e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 13.71528)
##
##     Null deviance: 2274.49  on 71  degrees of freedom
## Residual deviance:  960.07  on 70  degrees of freedom
## AIC: 396.83
##
## Number of Fisher Scoring iterations: 2
```

### Interpretation:

- The coefficient for `time_numeric` shows the average change per day.
- A significant positive slope suggests increasing symptoms over time.

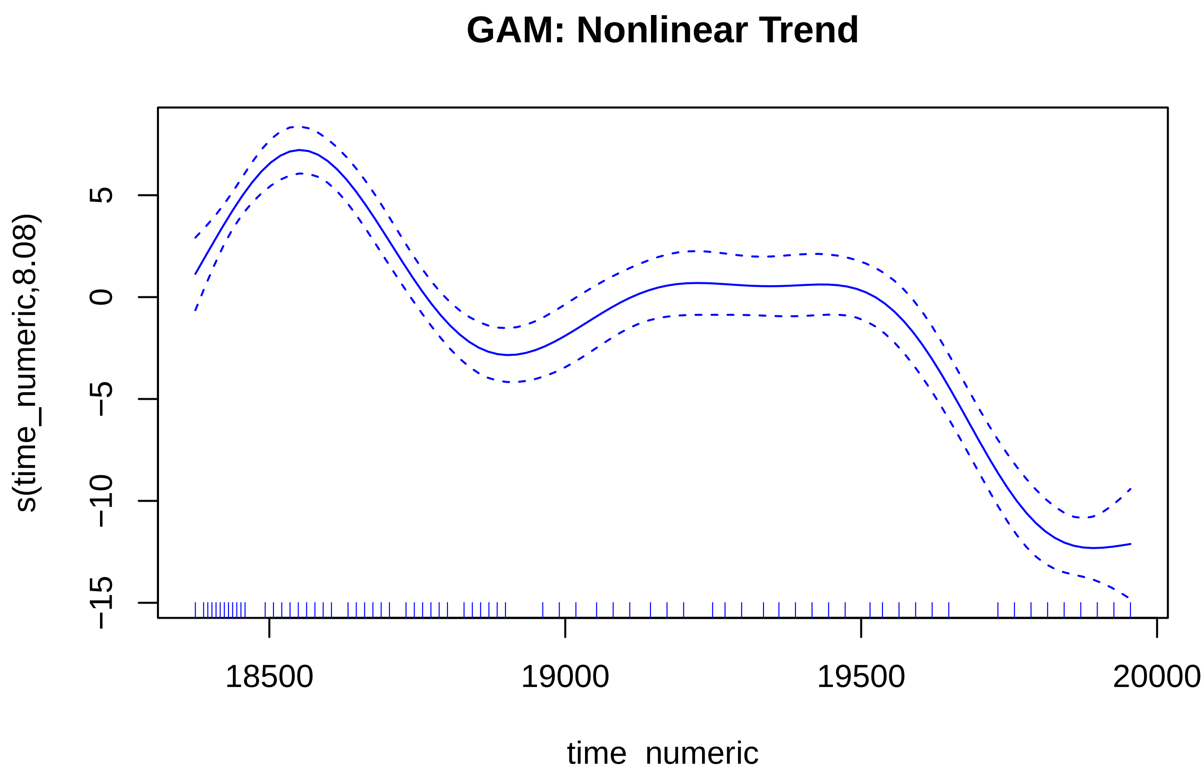
# Generalized Additive Model (GAM)

```
gam_model <- gam(value ~ s(time_numeric), data = df %>% filter(group == "National Estimate"))
summary(gam_model)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## value ~ s(time_numeric)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  32.9361     0.2424   135.9   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                   edf Ref.df     F p-value
## s(time_numeric) 8.076  8.772 53.49  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## R-sq.(adj) =  0.868   Deviance explained = 88.3%
## GCV = 4.8406  Scale est. = 4.2304    n = 72
```

```r
plot(gam_model, se = TRUE, col = "blue", main = "GAM: Nonlinear Trend")
```




### Interpretation:

- `s(time_numeric)` models smooth nonlinear trends.
- EDF > 1 confirms nonlinearity; p-value shows the trend is significant.

## Linear Mixed Effects Model (LMM)

```r
lmm_data <- df %>% filter(group %in% c("By Age", "By Race/Hispanic ethnicity"))
lmm <- lmer(value ~ time_numeric + (1|subgroup), data = lmm_data)
summary(lmm)
```

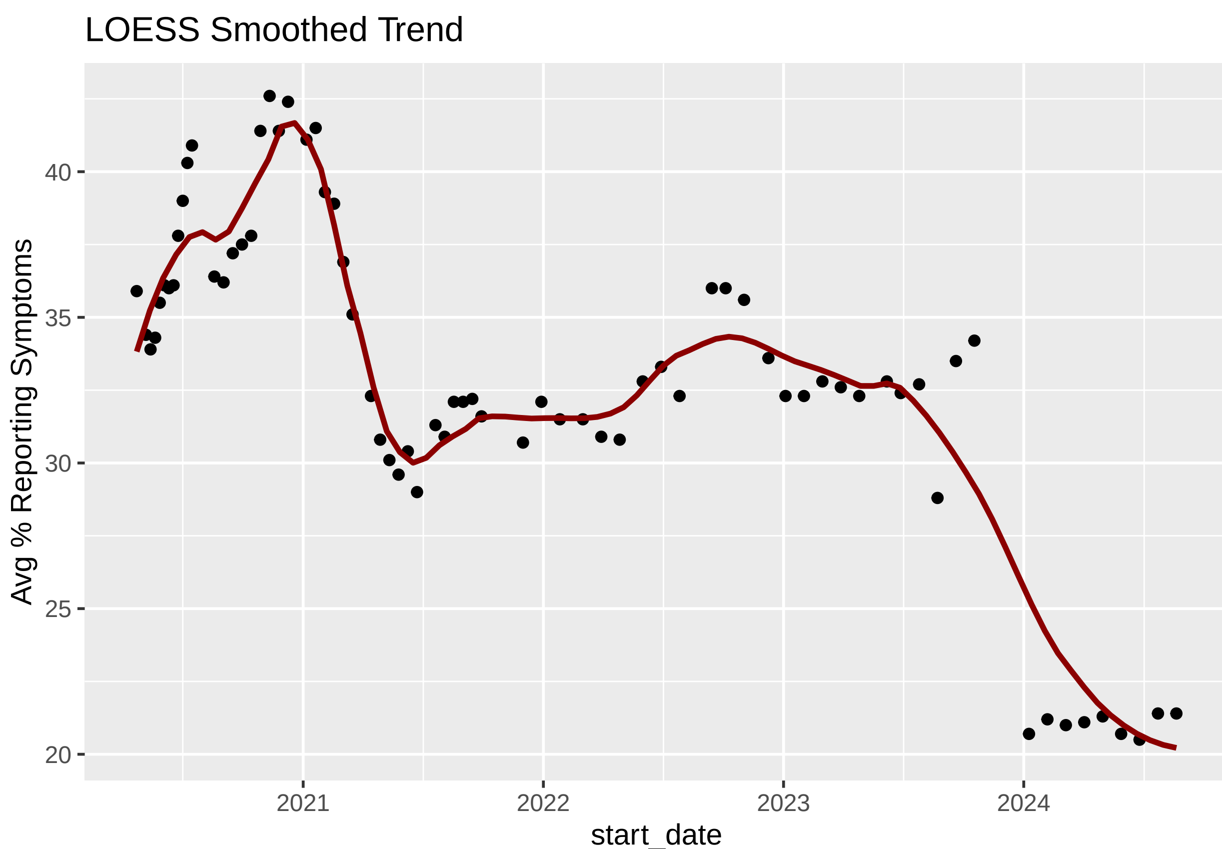
```
## Linear mixed model fit by REML ['lmerMod']
## Formula: value ~ time_numeric + (1 | subgroup)
##    Data: lmm_data
##
## REML criterion at convergence: 4931.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.74394 -0.79351 -0.05396  0.81619  2.55594
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subgroup (Intercept) 85.57    9.250
##  Residual             16.06    4.008
## Number of obs: 864, groups:  subgroup, 12
##
## Fixed effects:
##                Estimate Std. Error t value
## (Intercept)   2.018e+02  5.993e+00   33.66
## time_numeric -8.932e-03  2.823e-04  -31.64
##
## Correlation of Fixed Effects:
##             (Intr)
## time_numerc -0.895
```

### Interpretation:

- Fixed effect = average change over time.
- Random intercepts allow baseline variation across subgroups.

## Time Series Analysis (LOESS)

```
ts_data <- df %>%
  filter(group == "National Estimate") %>%
  group_by(start_date) %>%
  summarise(mean_value = mean(value))

ggplot(ts_data, aes(x = start_date, y = mean_value)) +
  geom_point() +
  geom_smooth(method = "loess", span = 0.3, se = FALSE, color = "darkred") +
  labs(title = "LOESS Smoothed Trend", y = "Avg % Reporting Symptoms")
```




### Interpretation:

- LOESS identifies short-term peaks and dips over time.

# MANOVA

```r
# Filter for Age and Race groups
manova_df <- df %>%
  filter(group %in% c("By Age", "By Race/Hispanic ethnicity"),
         !is.na(value)) %>%
  select(time_numeric, group, subgroup, value)

# Pivot to wide format: subgroup becomes columns
manova_wide <- manova_df %>%
  unite(varname, group, subgroup, sep = "_") %>%
  pivot_wider(names_from = varname, values_from = value) %>%
  drop_na()

# View shape
nrow(manova_wide)  # should be > 0
```

```
## [1] 72
```

```r
head(manova_wide)
```

```
## # A tibble: 6 x 13
##   time_numeric `By Age_18 - 29 years` `By Age_30 - 39 years`
##          <dbl>                  <dbl>                  <dbl>
## 1        18375                   46.8                   39.6
## 2        18389                   47.4                   39.3
## 3        18396                   47.7                   37.8
## 4        18403                   46.6                   39.5
## 5        18410                   49.3                   40.6
## 6        18417                   49.3                   41.5
## # i 10 more variables: `By Age_40 - 49 years` <dbl>,
## #   `By Age_50 - 59 years` <dbl>, `By Age_60 - 69 years` <dbl>,
## #   `By Age_70 - 79 years` <dbl>, `By Age_80 years and above` <dbl>,
## #   `By Race/Hispanic ethnicity_Hispanic or Latino` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic White, single race` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic Black, single race` <dbl>,
## #   `By Race/Hispanic ethnicity_Non-Hispanic Asian, single race` <dbl>, ...
```

```r
# Run MANOVA: test effect of time on the set of mental health outcomes by age/race groups
response_vars <- manova_wide %>% select(-time_numeric)
fit <- manova(as.matrix(response_vars) ~ time_numeric, data = manova_wide)

# Summary with Wilks' lambda
summary(fit, test = "Wilks")
```

```
##              Df   Wilks approx F num Df den Df    Pr(>F)
## time_numeric  1 0.15765   26.272     12     59 < 2.2e-16 ***
## Residuals    70
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Interpretation:

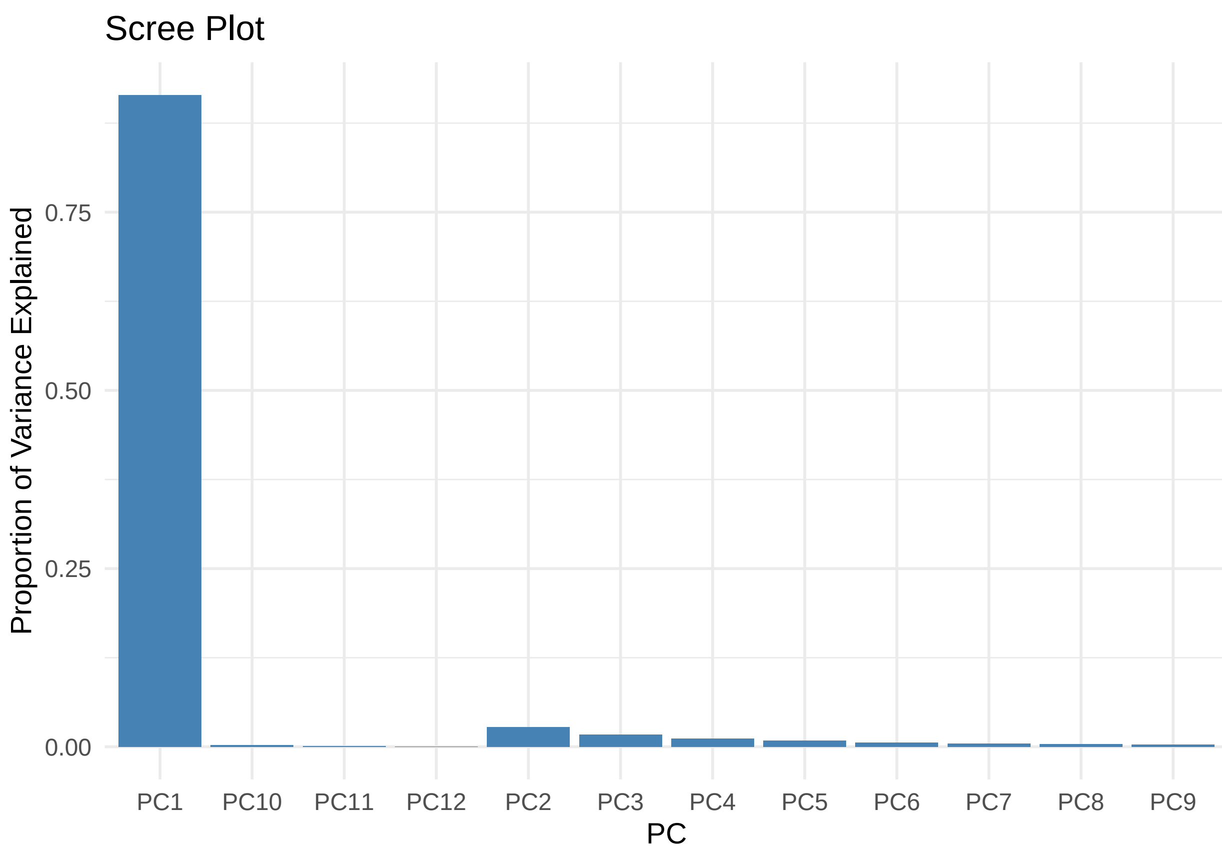
- Wilks' Lambda tests joint differences across time.
- Significant result indicates meaningful group-level changes.

## Principal Component Analysis (PCA)

```
# Prepare data for PCA
pca_data <- manova_wide %>%
  select(-time_numeric)  # only the mental health response variables

# Remove columns with zero variance (constant variables)
pca_data <- pca_data %>% select(where(~ sd(., na.rm = TRUE) > 0))

# Run PCA with scaling
pca_result <- prcomp(pca_data, scale. = TRUE)

# View summary of explained variance
summary(pca_result)
```

```
## Importance of components:
##                           PC1     PC2     PC3     PC4     PC5     PC6     PC7
## Standard deviation     3.3130 0.57494 0.45533 0.36997 0.32897 0.26309 0.23700
## Proportion of Variance 0.9146 0.02755 0.01728 0.01141 0.00902 0.00577 0.00468
## Cumulative Proportion  0.9146 0.94220 0.95948 0.97088 0.97990 0.98567 0.99035
##                            PC8     PC9    PC10   PC11    PC12
## Standard deviation     0.20530 0.18884 0.16212 0.1042 0.02954
## Proportion of Variance 0.00351 0.00297 0.00219 0.0009 0.00007
## Cumulative Proportion  0.99386 0.99683 0.99902 0.9999 1.00000
```

```
# Scree plot
library(ggplot2)
scree_df <- data.frame(PC = paste0("PC", 1:length(pca_result$sdev)),
                       Variance = pca_result$sdev^2 / sum(pca_result$sdev^2))
ggplot(scree_df, aes(x = PC, y = Variance)) +
  geom_col(fill = "steelblue") +
  theme_minimal() +
  labs(title = "Scree Plot", y = "Proportion of Variance Explained")
```



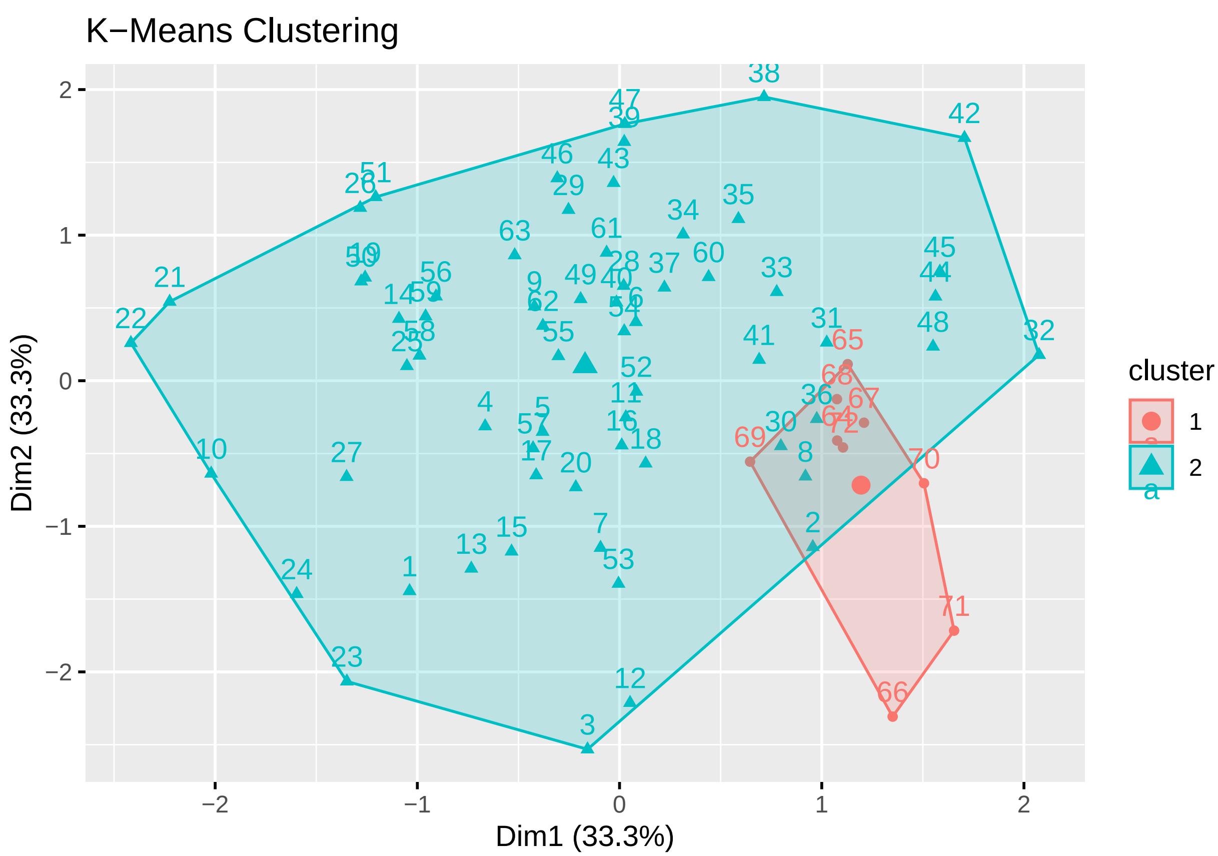

```
# Create a data frame of the first few principal components for clustering
pca_df <- as.data.frame(pca_result$x[, 1:3])  # Use top 3 PCs
```

### Interpretation:

- PCA reveals latent patterns in subgroup responses.
- PC1 usually captures general trend; others may reflect specific group contrasts.

## Clustering Analysis (K-Means)

```
clust_data <- scale(na.omit(pca_df))
k2 <- kmeans(clust_data, centers = 2, nstart = 25)
fviz_cluster(k2, data = clust_data, main = "K-Means Clustering")
```

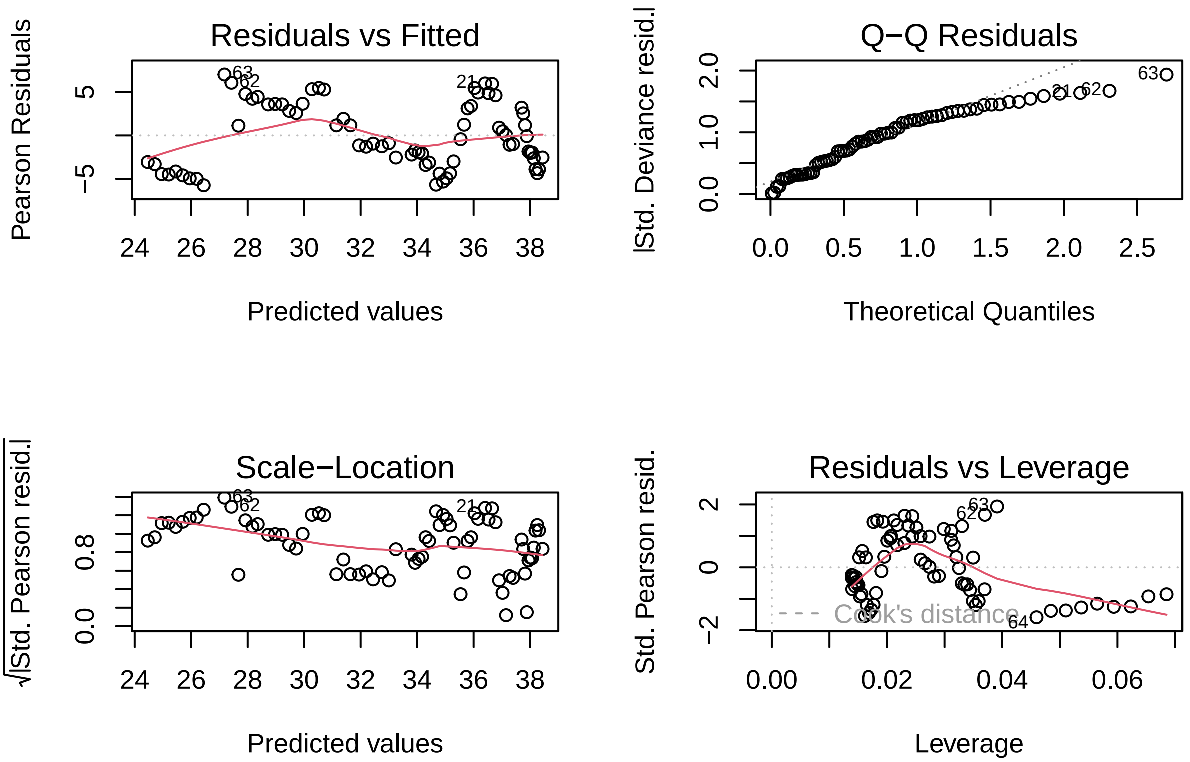



### Interpretation:

- Clustering groups similar subgroup response patterns.
- Useful for targeting interventions to specific demographic clusters.

## Model Diagnostics

```
par(mfrow = c(2, 2))
plot(glm_model)
```




```
#plot(gam_model, residuals = TRUE, pch = 20)
```

### Interpretation:

- Check assumptions: normality, homoscedasticity, outliers.
- GAM plots also show if smooths are well-fit.

## Conclusion

The analysis revealed complex, nonlinear trends in self-reported mental health outcomes during the pandemic. Generalized Additive Models captured smooth variation, while Mixed Effects Models showed heterogeneity across subgroups. MANOVA confirmed significant temporal changes by age and race. PCA and clustering exposed latent patterns and helped classify at-risk groups. These findings underscore the importance of advanced analytics for understanding large public health datasets.

## References


- CDC Household Pulse Survey: https://www.cdc.gov/nchs/covid19/pulse/mental-health.htm
- Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R.*
- Pinheiro & Bates (2000). *Mixed-Effects Models in S and S-PLUS.*
- Jolliffe, I. T., & Cadima, J. (2016). *Principal component analysis: a review and recent developments.*